

Brucellosis

Primary health care block 2025-2026

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- Case presentation

A doctor is struggling to diagnose a 40-year-old woman with flu-like illness. She complains of a fever that rises during the day and peaks after dinner, fatigue, spinal tenderness, and loss of appetite. Her lymph nodes are enlarged in the physical exam. The doctor has trouble narrowing down the possible etiologies until he hears that the woman tasted goat cheese at a local village a month before the onset of her symptoms.

- On physical examination, a temperature of 37.9°C, a pulse rate of 102 per minute, a respiratory rate of 24 per minute, and a blood pressure of 126/94 mm Hg were recorded.
- Laboratory results showed a hematocrit of 37.5% and a WBC count of 13,700/mm³, with 69% PMNs.
- glucose 69 mg/dL; Two blood cultures were negative for *B. abortus*, serology positive for IgM .
- The patient was treated with Tetracycline and rifampin for 4 weeks and fully recovered.
- Interpret ?????????

Brucellosis is a systemic bacterial zoonotic disease caused by the bacteria of the genus *Brucella*.

These bacteria are primarily passed among animals, and they cause disease in many different vertebrates.

Various *Brucella* species affect sheep, goats, cattle, deer, elk, pigs, dogs, and several other animals

B. abortus (cattle), *B. melitensis* and *B. ovis* (sheep and goats)

B. suis (pigs), *B. canis* (dogs)).

Humans become infected by contact with animals or animal products that are contaminated with these bacteria.

Brucellosis is predominantly an occupational disease affecting individuals who work with infected animals or their tissues.

* veterinarians, farmers, abattoir workers, butchers, lab staff

→ meat & milk

The following are the main ways that germs enter humans from animals:

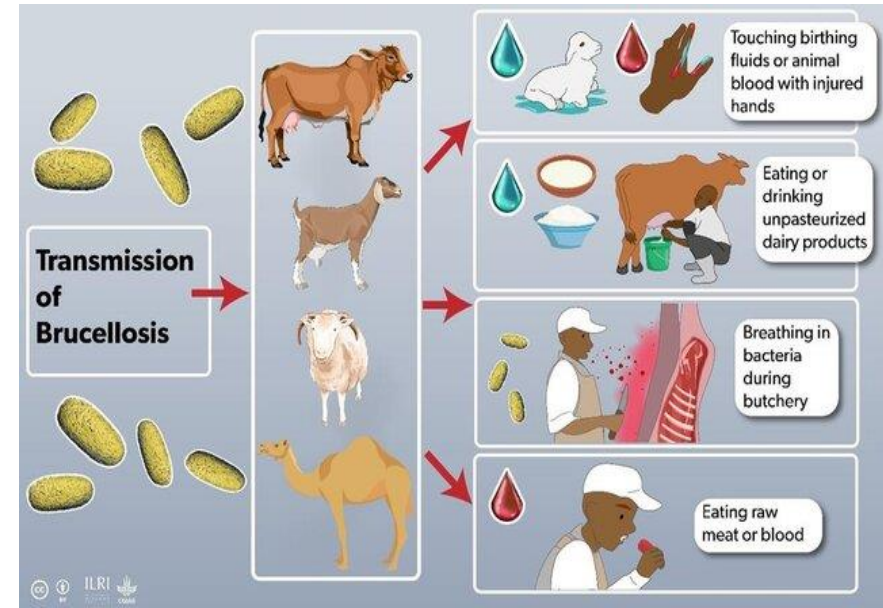
- Contacting the bodily fluids and blood of affected animals. Through a cut or other wound, bacteria from the blood, semen, or placenta of an infected animal might enter your circulation.

- individuals with compromised immune systems should avoid touching dogs that are known to be ill. *also other compromised animals*

- Consuming dairy products in their unprocessed state & unpasteurized milk, ice cream, butter, and cheeses that contain the bacteria. *unpasteurized milk*

The meat of infected animals that is uncooked or undercooked can potentially spread the bacterium.

- Breathing in tainted air. The bacteria Brucella spread easily through the air. The bacterium can be inhaled by workers in slaughterhouses, lab technicians, farmers, and hunters.



* can be transmitted by blood & bodily fluids, inhalation, GIT tract

Clinical presentation

Brucellosis presents as an acute, undulant, or chronic illness. symptoms may include chronic fatigue syndrome, depression, and arthritis

The incubation period is typically 5–30 days, but may be up to 6 months. *→ bc 11/19 Subclinical infection become symptomatic*

. Symptoms are nonspecific and flulike (malaise, fever, sweats, anorexia, GI symptoms, headache, and back pains) .

Symptom onset may be abrupt *→ acute* or insidious *→ gradual*. The infection may persist for several months without causing any symptoms. Person-to-person spread is rare, although mother-to-child and sexual transmission have been documented.

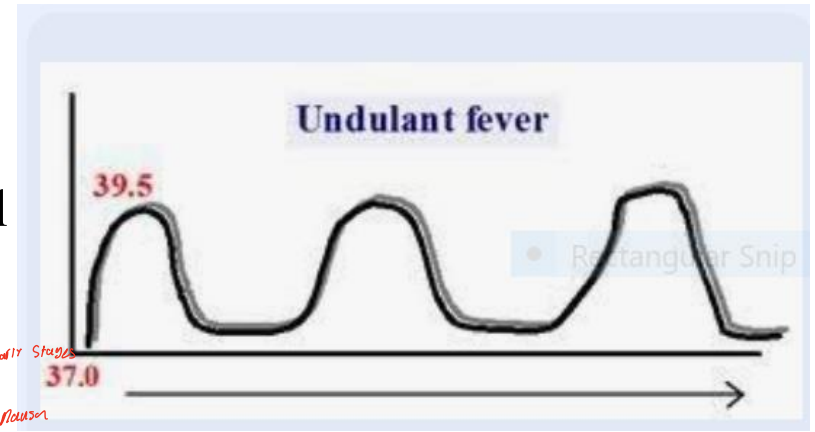
It can cause a range of symptoms that are similar to a flu-like illness and may include continued, intermittent or irregular fever of variable duration; headache; back pains, weakness; profuse sweating; chills; depression; and joint pain.

Recovery is usual, but disability *→ like back pain or Arthritis* is often pronounced. The case-fatality rate of untreated brucellosis is 2% or less.

Untreated, patients may develop an undulating pattern of fever (temperatures repeatedly rise then fall, hence the name “undulant fever,” the traditional name for brucellosis).

Manifestations of brucellosis may involve any of a variety of organ systems, including the GI tract and the skeletal, neurologic, cardiovascular, and pulmonary systems

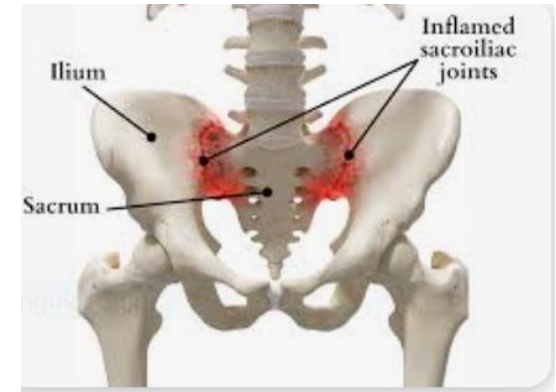
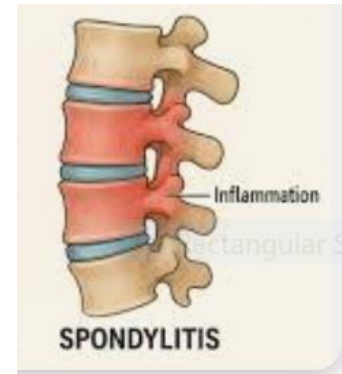
The symptoms of brucellosis may go away for a few weeks or months before coming again. Even after receiving therapy, some patients with persistent brucellosis continue to have symptoms for years.



typical symptoms are flu like + joint pain → musculoskeletal symptoms → common target

Long-term indicators and symptoms could consist of:

- Spinal bone arthritis (spondylitis) *→ often confused with disc prolapse*
- Frequent fevers
- Arthritis in the joints that link the pelvis and spine (sacroiliitis)
- Endocarditis, or inflammation of the heart's inner lining
- Irritation of the joints (arthritis)



Brucellae are Gram-negative coccobacilli; they are facultative intracellular parasites, multiplying mainly in monocyte-macrophage cells.

Their ability to resist phagocytosis is a major virulence determinant that enables them to cause disease.

Growth is slow and scanty.. colonies may appear after 14 days to 4 weeks. } *hence usually diagnosed by serological test (indirect agar)*

Brucella need enriched media such as blood, chocolate and MacConkey agars

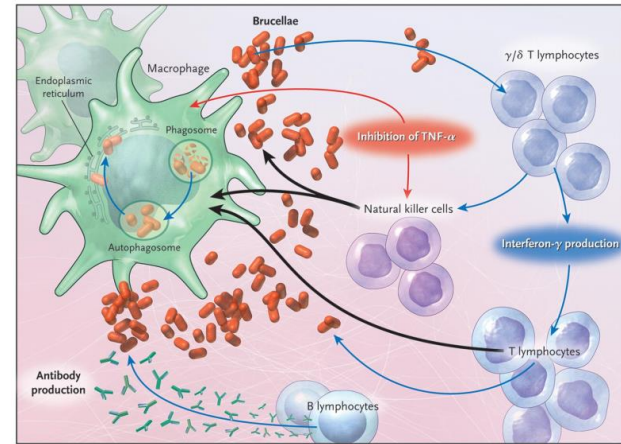


Pathogenesis :

Brucellae typically enter the body through cuts and abrasions in the skin or through the gastrointestinal (GI) tract. Drugs that decrease gastric acidity may increase the likelihood of transmission via the GI route. Inhalation of infected aerosols ^{→ 90%} can also lead to disease among workers. Once the organisms gain entry, they are transported via the lymphatic system to the regional lymph nodes, where they multiply. The organisms are then carried by the blood to organs that are involved in the reticuloendothelial system, including the liver, spleen, kidneys, bone marrow, and other lymph nodes.

[↳] in difficult cases used for diagnosis

↳ hence good to take CS for hepatomegaly, splenomegaly



Laboratory identification

Because the nonspecific symptoms may not point to a diagnosis of brucellosis, a detailed history is often crucial, including the patient's occupation, exposure to animals, travel to countries where brucella infection is prevalent, and ingestion of potentially contaminated foods..

Diagnosis is established by culturing the organism, usually from the blood, other body fluids or extract samples from bone marrow or tissues specimens.

Accurate identification of *Brucella* species relies on isolation of the pathogen from blood, bone marrow, or other tissues.

The success of culture-based detection varies according to disease stage, sample type, prior antimicrobial exposure, and culture technique used. Despite its limited sensitivity, culture remains the most definitive method of diagnosis

→ gold standard for
after infection

→ like CSF, synovial fluid

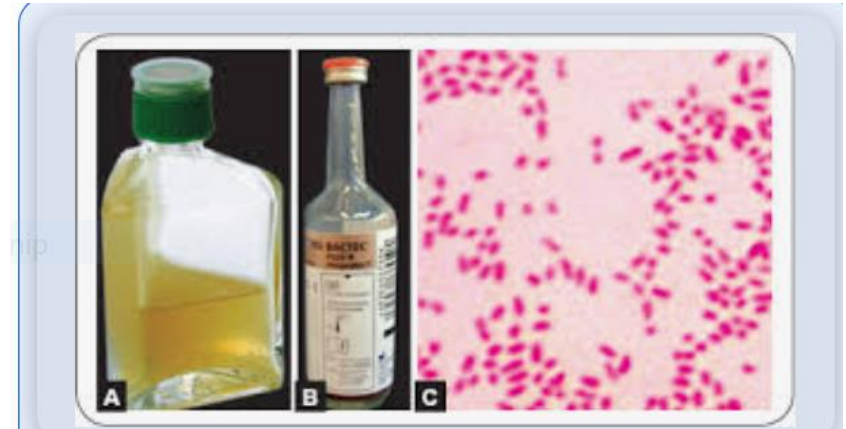
although very difficult in brucellosis

→ blood sample → better diagnosis

During ^{→ false -ve} early infection, the bacterial load in the bloodstream is typically low and may be missed if the sample size is insufficient. To maximize diagnostic yield, it is recommended that two or three separate peripheral blood cultures be obtained, ^{most during culture again 10 days later}

In severe cases, traditional ^{Manual} culture methods may require incubation for up to 7 days, whereas automated systems may detect growth within 5 days .

Within 24 h of infection, an estimated 25%–35% of patients may exhibit dissemination of *Brucella* beyond the bloodstream. Cultures may also be performed from bone marrow, urine, liver biopsies, ^{→ especially if not abscess} lymph nodes, and cerebrospinal fluid and ^{→ in immunocompromised} incubated at 35 °C in 5% CO₂ for up to 2 weeks ^{→ especially in elderly}

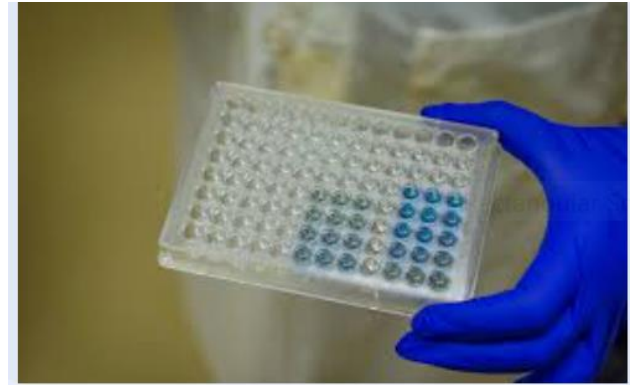
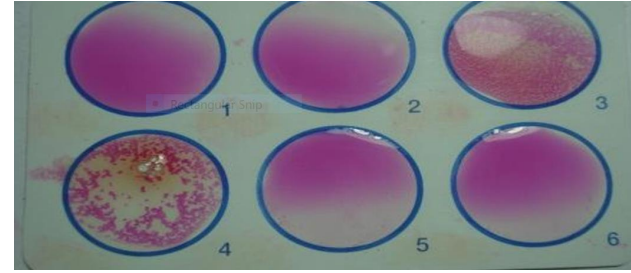


→ smaller & diagnosed
for this since faster than culture

Serological tests provided indirect evidence of infection: conventional testing uses antigens of *B. melitensis* and *B. abortus*, but serological responses do not completely distinguish between the two species. Serological assays, such as the Rose Bengal test and ELISA, offer rapid screening capabilities but can be limited in both sensitivity and specificity.

Most serological assays target antibodies against smooth LPS or cytosolic proteins. The immune response to smooth LPS—common to smooth *Brucella* species—results in sequential production of IgM (first week) and IgG1 (second week), followed by IgG2 and IgA (third week).

Molecular approaches, including PCR offer faster detection of *Brucella* DNA and are particularly valuable



Imaging technique ^{lost resource}

Echocardiography, bone scans, CT scans, MRIs, X-rays, ultrasounds, MRI,
Depending on which sections of your body are impacted. These take images of the
interior of your body to check for changes in your organs or bones.

* sometimes no fever no loss of appetite just joint pain

↳ distortion of bones



Brucellosis treatment regimens

→ Compliance detect → since harder later return } → often causing disease easy to diagnose but complicated treatment

Subsequent studies demonstrated that combination antibiotic therapy produced significantly better results than monotherapy, reducing relapse rates and improving overall efficacy

Today, the cornerstone of brucellosis treatment remains antimicrobial therapy, particularly the use of dual antibiotics such as doxycycline (100 mg twice daily for 6 weeks) in combination with either streptomycin (1 g intramuscularly daily for 2–3 weeks) or rifampicin (600–900 mg daily for 6 weeks)

→ Protein inhibitor
→ cause deafness
→ can't use in children
→ basicic inhibition

→ cause liver problem

* treatment? is always in combination and atleast 9 week → due to risk of reactivation (recurrence) (latency)

Epidemiology:

Brucellosis is a chronic, lifelong infection in animals.

Organisms localize in reproductive organs (male and female) and are shed in large numbers in milk, urine, the placenta, and other tissues discharged during delivery or spontaneous abortion. The primary manifestations of infection in animals are sterility and abortion.

Transmission to humans characteristically occurs as a result of either direct contact with infected animal tissue or ingestion of unpasteurized milk or milk products

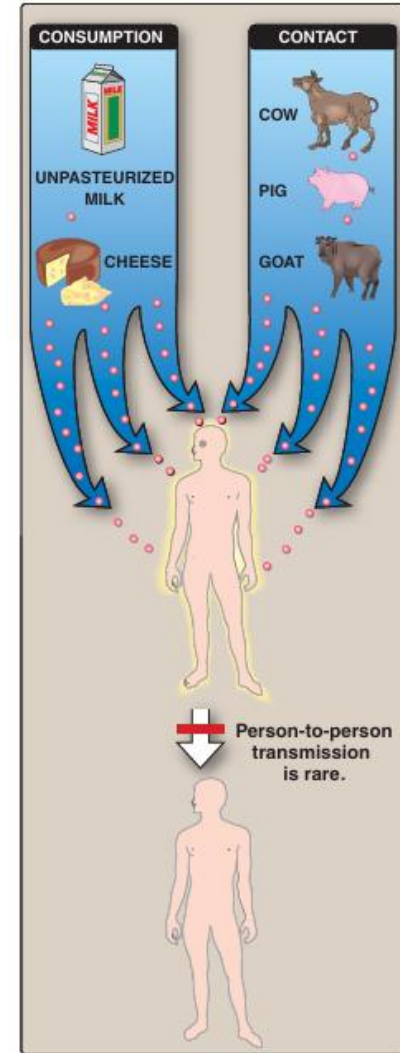
Prevention In most developed countries

1/ *B. abortus* has been eradicated from cattle (through vaccination, test and slaughter of infected herds, and by pasteurization of milk products) and is now a rare cause of disease.

boiling (70% for 15 min)
(100% for 5-10 min)

some lines of diagnostic test

cause animal tissues abortion



2/ Avoid consuming unpasteurized milk products or drinking them.

3/ Wear the proper protective clothing when handling animal tissues and animals. This might be an apron, or gloves. Those who work in slaughterhouses or medical laboratories, as well as hunters, farmers, veterinarians, and butchers, need to exercise extra caution.

3/ Handle all specimens ^{as it's infectious so to be safe follow these steps} under the proper biosafety circumstances if you operate in a laboratory. Additionally, slaughterhouses must adhere to safety precautions, including wearing protective gear and keeping the killing floor isolated from other processing areas.

Inadequate biosafety protocols, particularly in resource-limited settings, increase the risk of laboratory-acquired infections ^{so Brucellosis is}

2. antibiotic

3. occupational acquired

4. laboratory acquire

4/ Always wash your hands, as well as the surfaces and utensils you used to create your food and cook meat to safe temperatures.

Self study

Reinterpret the case according to the lecture and find any defects in the management line